

VISUALIZATION

Data

Overview

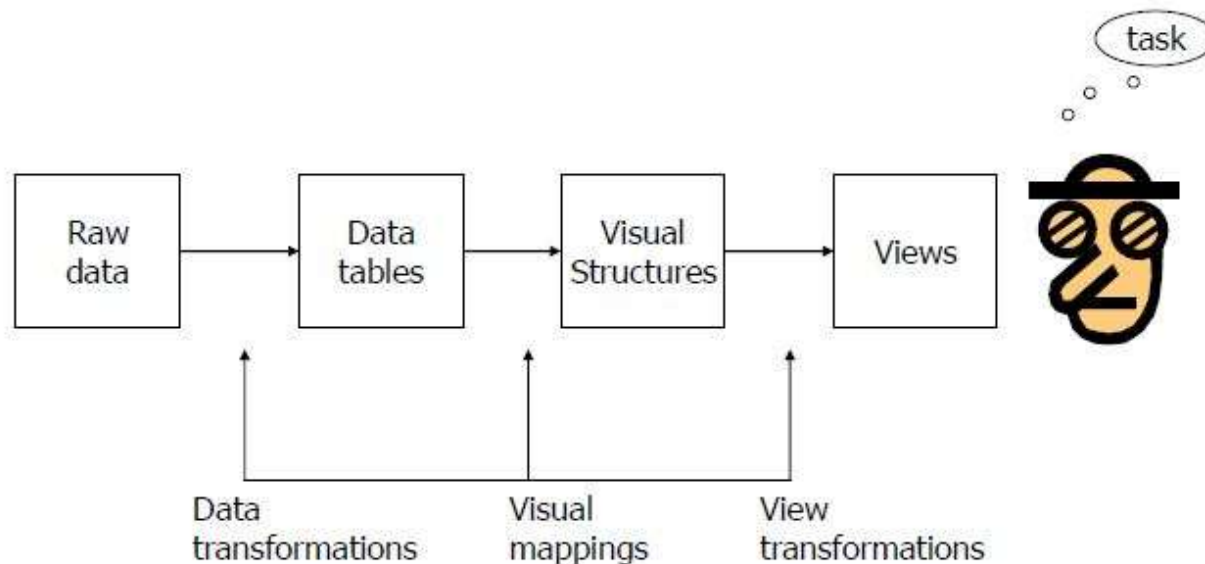
- Data
- Attribute Types
- Dataset Types

Overview

- Data
- Attribute Types
- Dataset Types

Visualization

- Visualization is the process that **transforms** (abstract) **data** into **interactive graphical representations** for the purpose of **exploration, confirmation, or presentation**
- **Goal:**
 - Record - Preserve information
 - Analyze - Reason about data
 - Present - Convey information



Data

- Data is taken from and/or representing some phenomena from the world
- Data Types are structural or mathematical interpretation of data
 - Fundamental unit
 - Item, Link, Position, Grid
 - Different from data types in programming!

Items & Attributes

Item: individual entity, discrete

e.g., Patient, Car, Stock, City

Attribute: measured,
observed, logged property

e.g., Patient: height, blood
pressure; Car: horsepower, make

Item: Person Attributes

ID	Name	Age	Shirt Size	Favorite Fruit
1	Amy	8	S	Apple
2	Basil	7	S	Pear
3	Clara	9	M	Durian
4	Desmond	13	L	Elderberry
5	Ernest	12	L	Peach
6	Fanny	10	S	Lychee
7	George	9	M	Orange
8	Hector	8	L	Loquat
9	Ida	10	M	Pear
10	Amy	12	M	Orange

Cell

Other Data Types

- Links/Relations
 - Express relationship between two items
 - Friendship on Facebook, Interaction between proteins
- Positions
 - Spatial data -> location in 2D or 3D
 - Pixels in photo, Voxels in MRI scan, latitude/longitude
- Grids
 - Sampling strategy for continuous data

Data Semantics

ID	Name	Age	Shirt Size	Favorite Fruit
1	Amy	8	S	Apple
2	Basil	7	S	Pear
3	Clara	9	M	Durian
4	Desmond	13	L	Elderberry
5	Ernest	12	L	Peach
6	Fanny	10	S	Lychee
7	George	9	M	Orange
8	Hector	8	L	Loquat
9	Ida	10	M	Pear
10	Amy	12	M	Orange

- Basil, 7, S, Pear
- What does it mean?
- **Semantics:** real world meaning
- Name? City? Fruit? Height? Age? Day of Month?
- Metadata: Descriptive information about the data
 - Is utilized to associate Semantics

Metadata

- Descriptive information about the data
 - Might be something as simple as the type of a variable, or could be more complex
 - For times when the table itself just isn't enough
 - Example: if variable1 is "I", then variable3 can only be 3, 7 or 16

Example - Item

◇	A	B	C	S	T	U
1	Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
2	3	10/14/06	5-Low	Large Box	0.8	10/21/06
3	6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
4	32	7/16/07	2-High	Small Pack	0.79	7/17/07
5	32	7/16/07	2-High	Jumbo Box	0.72	7/17/07
6	32	7/16/07	2-High	Medium Box	0.6	7/18/07
7	32	7/16/07	2-High	Medium Box	0.65	7/18/07
8	35	10/23/07	4-Not Specified	Wrap Bag		10/24/07
9	35	10/23/07	4-Not Specified	Small Box		10/25/07
10	36	11/3/07	1-Urgent	Small Box		11/3/07
11	65	3/18/07	1-Urgent	Small Pack		3/19/07
12	66	1/20/05	5-Low	Wrap Bag		1/20/05
13	69	6/4/05	4-Not Specified	Small Pack		6/6/05
14	69	6/4/05	4-Not Specified	Wrap Bag		6/6/05
15	70	12/18/06	5-Low	Small Box		12/23/06
16	70	12/18/06	5-Low	Wrap Bag		12/23/06
17	96	4/17/05	2-High	Small Box	0.55	4/19/05
18	97	1/29/06	3-Medium	Small Box	0.38	1/30/06
19	129	11/19/08	5-Low	Small Box	0.37	11/28/08
20	130	5/8/08	2-High	Small Box	0.37	5/9/08
21	130	5/8/08	2-High	Medium Box	0.38	5/10/08
22	130	5/8/08	2-High	Small Box	0.6	5/11/08
23	132	6/11/06	3-Medium	Medium Box	0.6	6/12/06
24	132	6/11/06	3-Medium	Jumbo Box	0.69	6/14/06
25	134	5/1/08	4-Not Specified	Large Box	0.82	5/3/08
26	135	10/21/07	4-Not Specified	Small Pack	0.64	10/23/07
27	166	9/12/07	2-High	Small Box	0.55	9/14/07
28	193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
29	194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

Item/Element/
(Independent)
Variable

Example - Attribute

	A	B	C	S	T	U
1	Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
2	3	10/14/06	5-Low	Large Box	0.8	10/21/06
3	6	2/21/08	4-Not Specified	Small Pack		2/22/08
4	32	7/16/07	2-High	Small Pack		7/17/07
5	32	7/16/07	2-High	Jumbo Box		7/17/07
6	32	7/16/07	2-High	Medium Box		7/18/07
7	32	7/16/07	2-High	Medium Box		7/18/07
8	35	10/23/07	4-Not Specified	Wrap Bag		10/24/07
9	35	10/23/07	4-Not Specified	Small Box		10/25/07
10	36	11/3/07	1-Urgent	Small Box		11/3/07
11	65	3/18/07	1-Urgent	Small Pack		3/19/07
12	66	1/20/05	5-Low	Wrap Bag		1/20/05
13	69	6/4/05	4-Not Specified	Small Pack	0.44	6/6/05
14	69	6/4/05	4-Not Specified	Wrap Bag	0.6	6/6/05
15	70	12/18/06	5-Low	Small Box	0.59	12/23/06
16	70	12/18/06	5-Low	Wrap Bag	0.82	12/23/06
17	96	4/17/05	2-High	Small Box	0.55	4/19/05
18	97	1/29/06	3-Medium	Small Box	0.38	1/30/06
19	129	11/19/08	5-Low	Small Box	0.37	11/28/08
20	130	5/8/08	2-High	Small Box	0.37	5/9/08
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25	134	5/1/08	4-Not Specified	Large Box	0.82	5/3/08
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27	166	9/12/07	2-High	Small Box	0.55	9/14/07
28	193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
29	194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

Attribute/
Dimension/
(Dependent)
Variable/
Feature

Example - Semantics

	A	B	C	S	T	U
1	Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
2	3	10/14/06	5-Low	Large Box	0.8	10/21/06
3	6	2/21/08	4-Not Specified	Small Pack	0.5	2/22/08
4	32	7/16/07	2-High	Small Pack	0.9	7/17/07
5	32	7/16/07	2-High	Jumbo Box	0.72	7/17/07
6	32	7/16/07	2-High	Medium Box	0.6	7/18/07
7	32	7/16/07	2-High	Medium Box	0.65	7/18/07
8	35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
9	35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
10	36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
11	65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
12	66	1/20/05	5-Low	Wrap Bag	0.56	1/20/05
13	69	6/4/05	4-Not Specified	Small Pack	0.44	6/6/05
14	69	6/4/05	4-Not Specified	Wrap Bag	0.6	6/6/05
15	70	12/18/06	5-Low	Small Box	0.59	12/23/06
16	70	12/18/06	5-Low	Wrap Bag	0.82	12/23/06
17	96	4/17/05	2-High	Small Box	0.55	4/19/05
18	97	1/29/06	3-Medium	Small Box	0.38	1/30/06
19	129	11/19/08	5-Low	Small Box	0.37	11/28/08
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27	166	9/12/07	2-High	Small Box	0.55	9/14/07
28	193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
29	194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

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- Data
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- Dataset Types

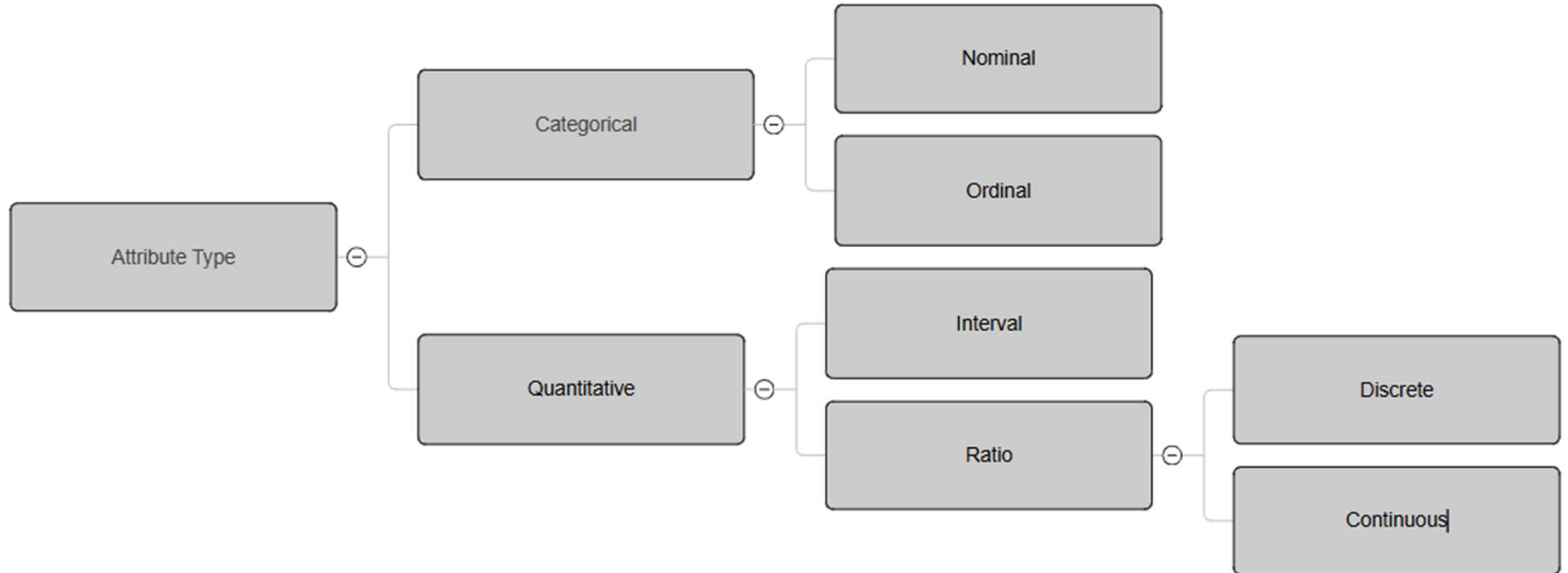
Attribute Types - Qualitative

- **Categorical (Qualitative):** A categorical (also called qualitative) attribute can take on one of a limited and usually fixed number of possible values. It is of two types:
 - **Nominal:** Attributes that do not have any order. Example: Gender, Car model, Fruit, Movie genres, etc.
 - **Ordinal:** Attributes that have some order. Example: Grade, Shirt size, Priority, etc.

Attribute Types - Quantitative

- **Quantitative:** A numerical attribute. It can be of various types:
 - **Interval:** A numerical attribute that does not have any true zero or the location of zero is arbitrary. Example: Date, Latitude, Longitude
 - **Ratio:** A numerical attribute that has a zero where there is nothing of the measured data. It can be of two types:
 - **Discrete:** Has a limited number of values and can be countable. Generally, these are whole numbers. Example: Goals scored in a football match, number of students in a class, etc.
 - **Continuous:** Data that falls in a continuous sequence and is not countable. Example: Temperature, Weight, etc

Attribute Types (Summary)



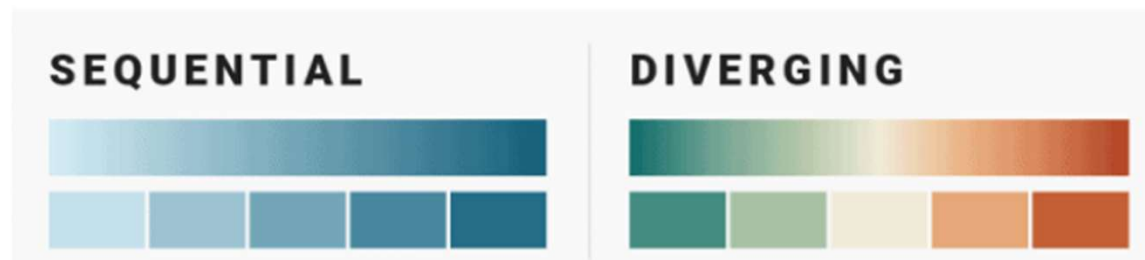
Operations

- Nominal (labels)
 - Operations: $=$, $\neq$
- Ordinal (ordered)
 - Operations: $=$, $\neq$, $>$, $<$
- Interval (location of zero arbitrary)
 - Operations: $=$, $\neq$, $>$, $<$, $+$, $-$ (distance)
- Ratio (zero fixed)
 - Operations: $=$, $\neq$, $>$, $<$, $+$, $-$, $\times$, $\div$ (proportions)

Sequential vs Diverging data

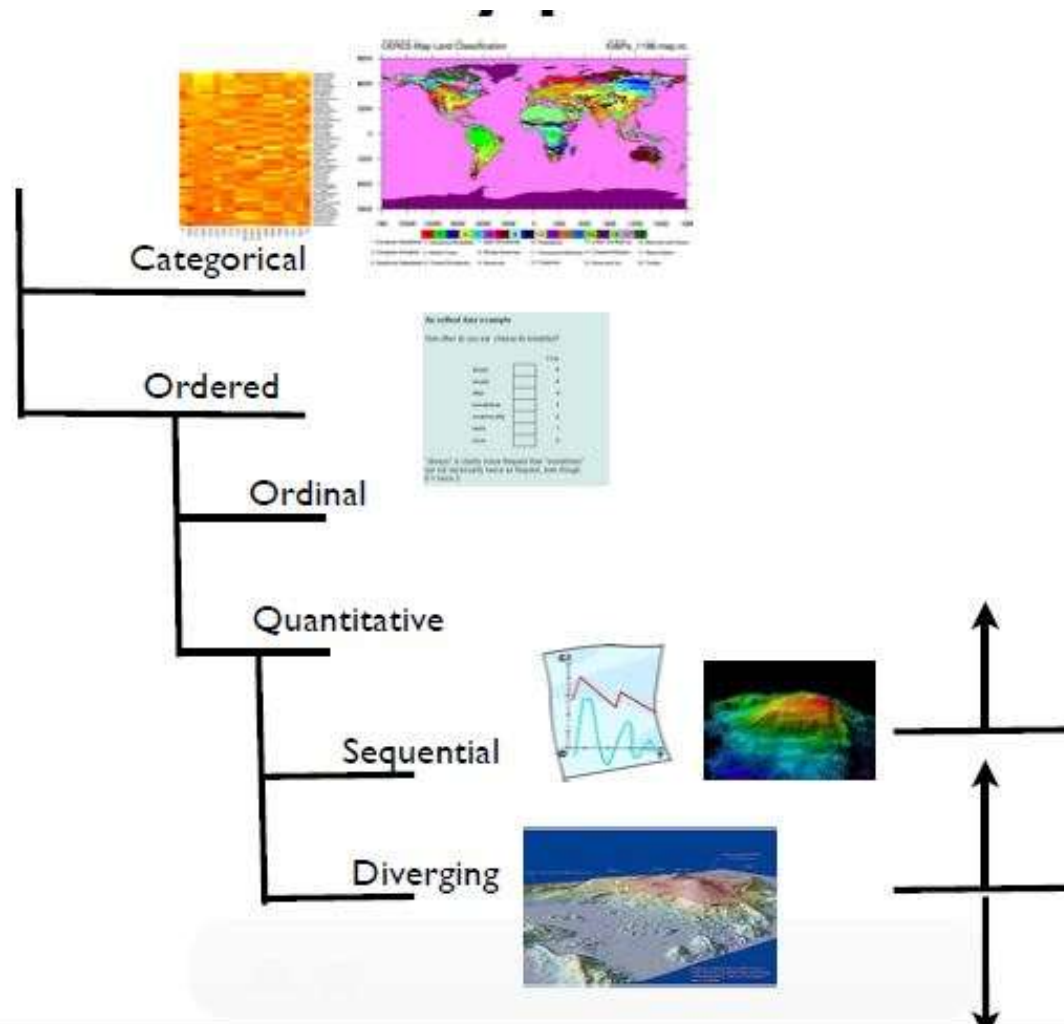
- Sequential:
 - Homogeneous from min to max
 - # people in countries
- Diverging:
 - Two or multiple sequences that meet
 - Elevation dataset: above sea level & below sea level

Sometimes, a single attribute can be represented either as sequential or diverging.



Sequential and Diverging Color Scales

Attribute Types (Summary)



Example

ID	NAME	DoB	GENDER	MARKS %	GRADE
1	Tom	6_2_2010	M	76.8	C
2	Cynthia	5_8_2009	F	92.1	A
3	Rajiv	12_12_2008	M	94.3	A

Name: Categorical - Nominal

DoB: Quantitative - Interval

Gender: Categorical - Nominal

Marks %: Quantitative – Ratio – Continuous

Grade: Categorical – Ordinal

Overview

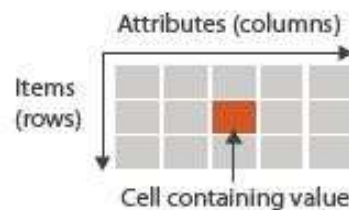
- Data
- Attribute Types
- Dataset Types

Dataset Types

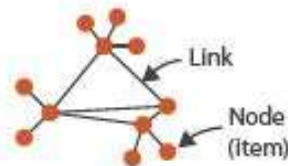
- Data types combine to form Dataset Types

➔ Dataset Types

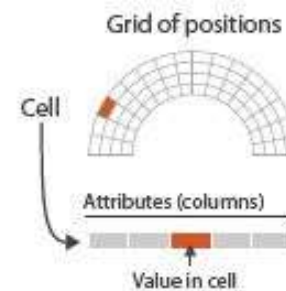
➔ Tables



➔ Networks



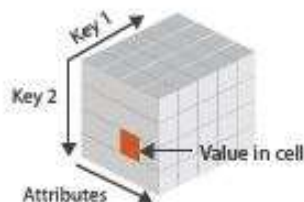
➔ Fields (Continuous)



➔ Geometry (Spatial)



➔ Multidimensional Table



➔ Trees



Tables

Flat Table

one item per row

each column is attribute

unique (implicit) **key**

no duplicates

Multidimensional Table

indexing based on multiple keys

		Attributes			
Keys		Values			
Item	ID	Name	Age	Shirt Size	Favorite Fruit
	1	Amy	8	S	Apple
	2	Basil	7	S	Pear
	3	Clara	9	M	Durian
	4	Desmond	13	L	Elderberry
	5	Ernest	12	L	Peach
	6	Fanny	10	S	Lychee
	7	George	9	M	Orange
	8	Hector	8	L	Loquat
	9	Ida	10	M	Pear
	10	Amy	12	M	Orange

2 dimensional Table

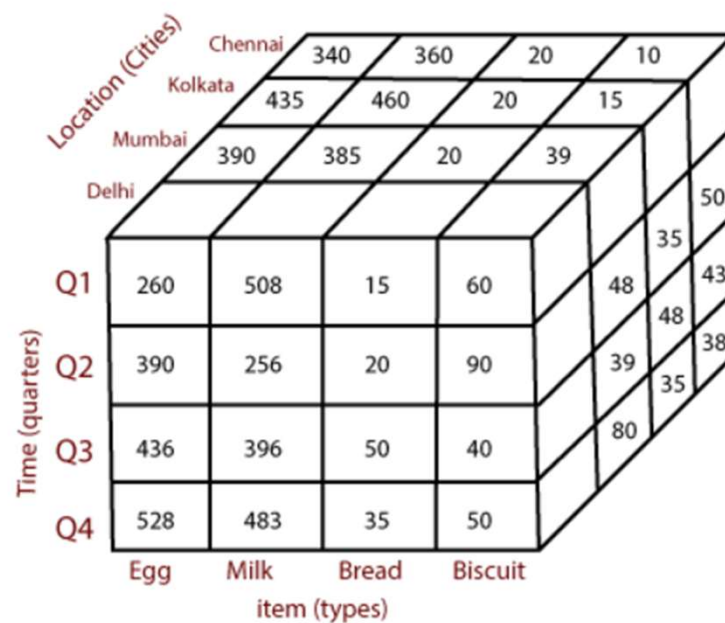
	A	B	C	D	E
1	#1.2				
2	1500	529			
3	GeneName	DESCRIPTION	TCGA-02-0001-01C-01R-0177-01	TCGA-02-0003-01A-01R-0177-01	TCGA-02-0004-01A-01R-0298-01
4	LTF	LTF	-1.265728057	2.377012066	4.123979585
5	POSTN	POSTN	2.662411805	3.932400324	5.031585377
6	TMSL8	TMSL8	-3.082217838	-2.243148513	-0.02313681
7	HLA-DQA1	HLA-DQA1	-1.739664398	4.577962344	3.127744964
8	RP11-35N6.1	RP11-35N6.1	-3.346352968	-2.895400157	-3.473035067
9	STMN2	STMN2	-2.578511106	-3.051605144	-1.729892888
10	DCX	DCX	-2.26078976	-2.529795801	-2.844966278
11	AGXT2L1	AGXT2L1	-2.639493611	-3.113204863	-0.403975027
12	IL13RA2	IL13RA2	-2.93596915	-1.873600916	2.976256911
13	SLN	SLN	-2.466718221	-2.208406749	1.025827904
14	MEOX2	MEOX2	-2.395054066	-1.062676046	1.783235317
15	COL11A1	COL11A1	1.211934832	-0.399392588	4.733608974
16	NNMT	NNMT	0.703745164	0.664082419	3.069030715
17	F13A1	F13A1	-0.224094042	2.222197544	1.171354775
18	CXCL14	CXCL14	-3.1309694	-1.395056071	2.569540659
19	MBP	MBP	-1.906390566	-2.037626447	-2.935744906
20	TF	TF	-4.334123292	-4.680680246	-2.975788866
21	KCND2	KCND2	-1.777692395	-2.100362021	-1.996306032
22	GABRB1	GABRB1	-2.214760175	-3.022654105	-3.185499425

Keys: Patients

Keys: Genes

3 dimensional Table

	Location="Chennai"				Location="Kolkata"				Location="Mumbai"				Location="Delhi"			
	item				item				item				item			
Time	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit
Q1	340	360	20	10	435	460	20	15	390	385	20	39	260	508	15	60
Q2	490	490	16	50	389	385	45	35	463	366	25	48	390	256	20	90
Q3	680	583	46	43	684	490	39	48	568	594	36	39	436	396	50	40
Q4	535	694	39	38	335	365	83	35	338	484	48	80	528	483	35	50



Represented as data cube

Variables

- Data sets of dimensions 1, 2, 3 are common
- Number of variables/attributes per class/item
 - 1: Uni-variate data
 - 2: Bi-variate data
 - 3: Tri-variate data
 - >3: Hyper-variate data
- Although multidimensional and multivariate data are closely related and often convertible, they are conceptually distinct.
- Multidimensional data emphasizes multiple keys or indexing dimensions, whereas multivariate data emphasizes multiple measured attributes per item.
- In practice, visualization systems often treat them similarly, which can blur the distinction.

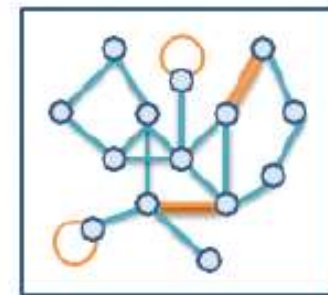
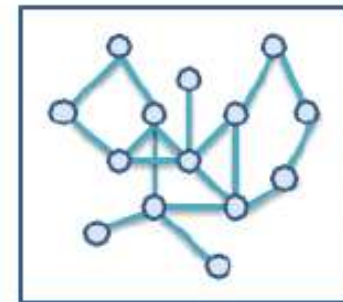
Data Table

- Items (*cases*) have attributes (*variables*)
- Function $f(\text{Case}_i) = \langle \text{Value}_{i1}, \text{Value}_{i2}, \dots \rangle$

D i m e n s i o n s		Case ₁	Case ₂	Case ₃	...
	Variable ₁	Value ₁₁	Value ₂₁	Value ₃₁	
	Variable ₂	Value ₁₂	Value ₂₂	Value ₃₂	
	Variable ₃	Value ₁₃	Value ₂₃	Value ₃₃	
	...				

Graphs/Networks

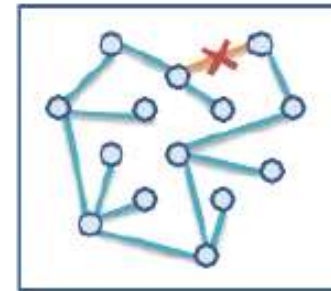
- A graph $G(V,E)$ consists of a set of **vertices (nodes)** V and a set of **edges (links)** E connecting these vertices.
- A simple graph is a graph which contains:
 - No multi-edges
 - No loops



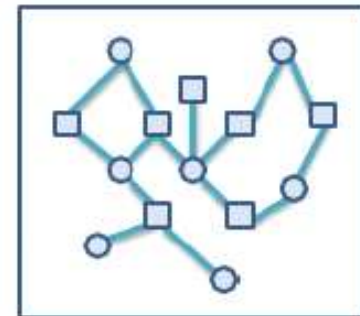
Not a simple graph!
→ A **general graph**

Special Graphs

- A **tree/hierarchy** is a graph with *no cycles*
- A **directed graph** (digraph) is a graph that distinguishes between edges $A \rightarrow B$ and $A \leftarrow B$
- A **bipartite graph** has vertices that can be partitioned into two independent sets



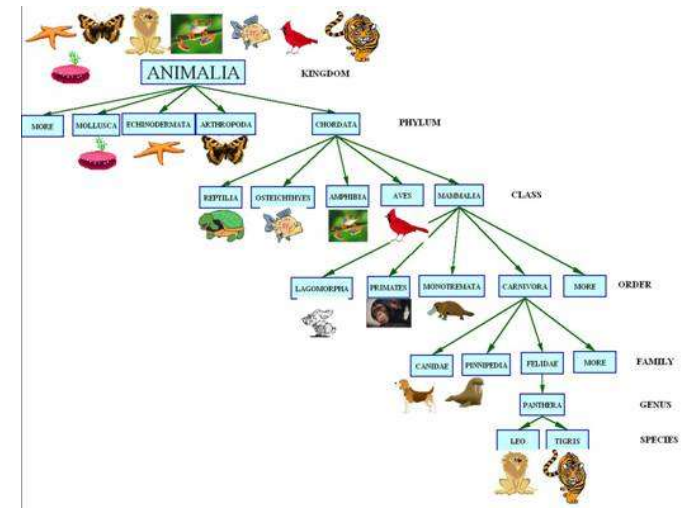
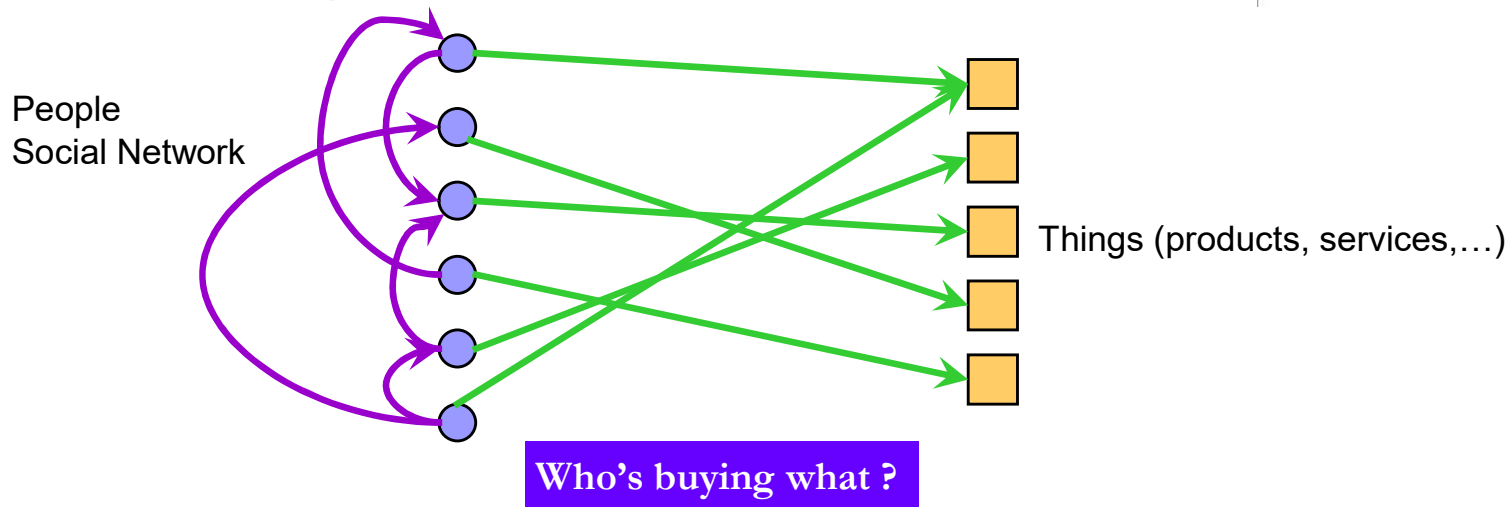
Tree



Bipartite Graph

Graphs in Real World

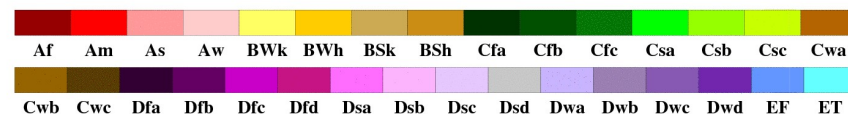
- Graph: WWW, Social Networks
- Tree: Org structures, Classifications
- Bipartite graph



Spatial Data

World Map of Köppen–Geiger Climate Classification

observed using CRU TS 2.1 temperature and GPCC Full v4 precipitation data, period 1976 to 2000



Main climates

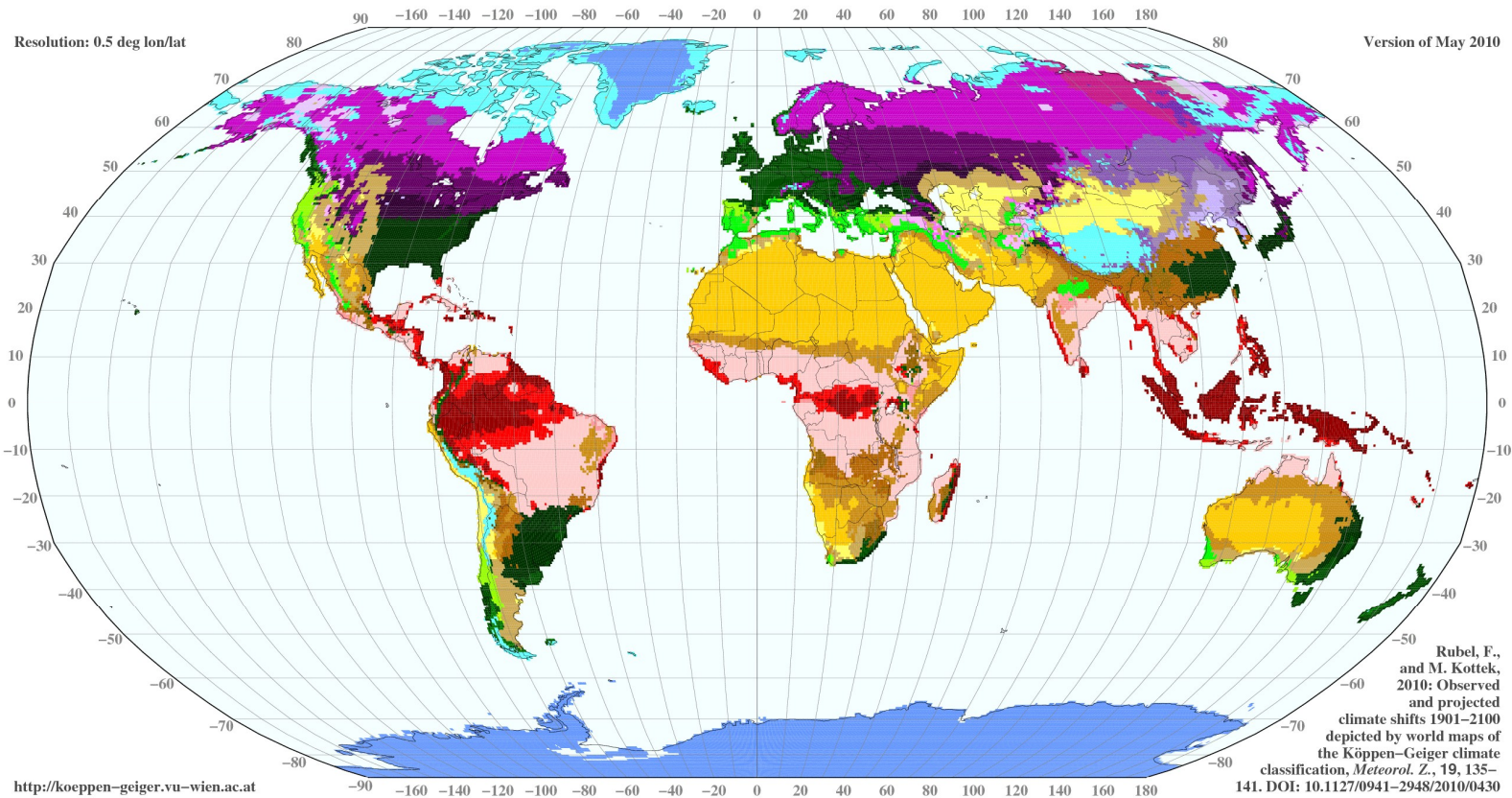
A: equatorial
B: arid
C: warm temperate
D: snow
E: polar

Precipitation

W: desert
S: steppe
f: fully humid
s: summer dry
w: winter dry
m: monsoonal

Temperature

h: hot arid
k: cold arid
a: hot summer
b: warm summer
c: cool summer
d: extremely continental
F: polar frost
T: polar tundra



Fields

Attribute values associated with cells

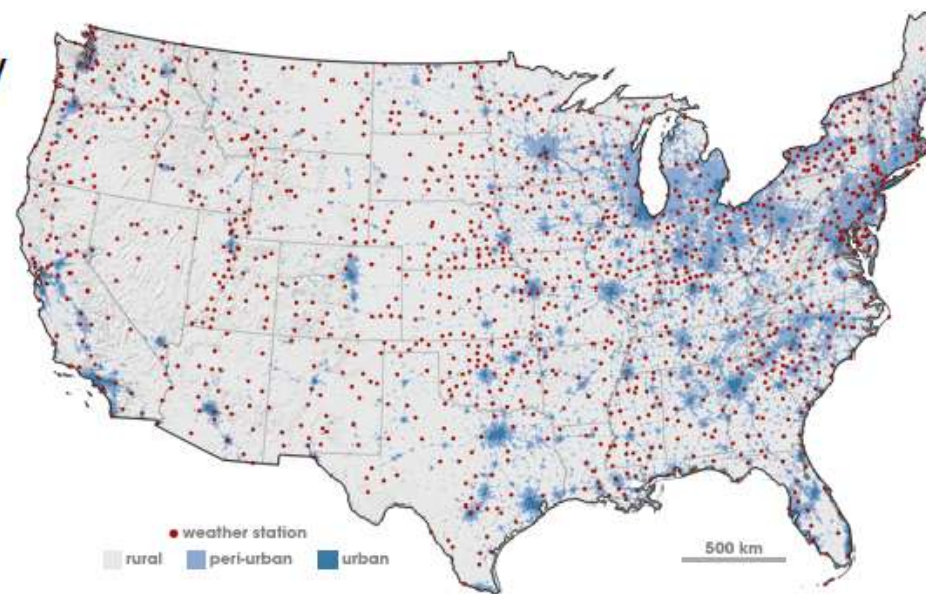
Cell contains data from continuous domain

Temperature, pressure, wind velocity

Measured or simulated

Sampling & Interpolation

Signal processing & stats

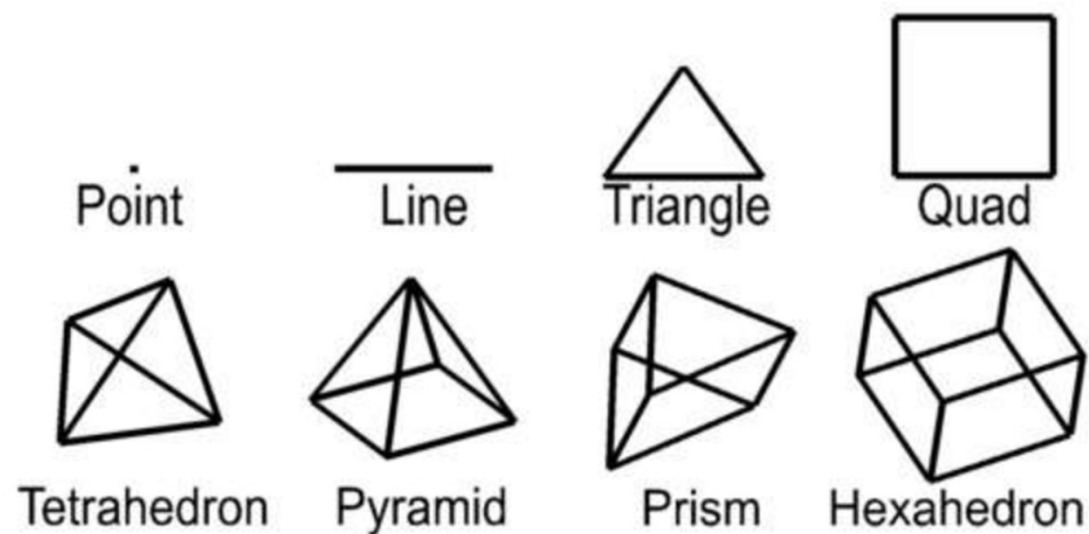


Grids

- Unless sparse measured data are mapped to a grid, computation of areas or volumes is not possible.
 - Further, the techniques available for visualizing the data will be very limited.
 - Therefore, a grid is created into which the data will be interpolated and extrapolated.
-
- A grid is defined as a collection of nodes and cells.
 - Nodes are points in two or three-dimensions with coordinates and usually one or more data values.
 - Cell refer to geometric objects.
 - The cell type and the nodes that comprise their vertices define these objects.

Cells

Cell Type	Number of Nodes	Dimensionality
Point	1	0
Line	2	1
Triangle	3	2
Quadrilateral	4	2
Tetrahedron	4	3
Pyramid	5	3
Prism	6	3
Hexahedron	8	3



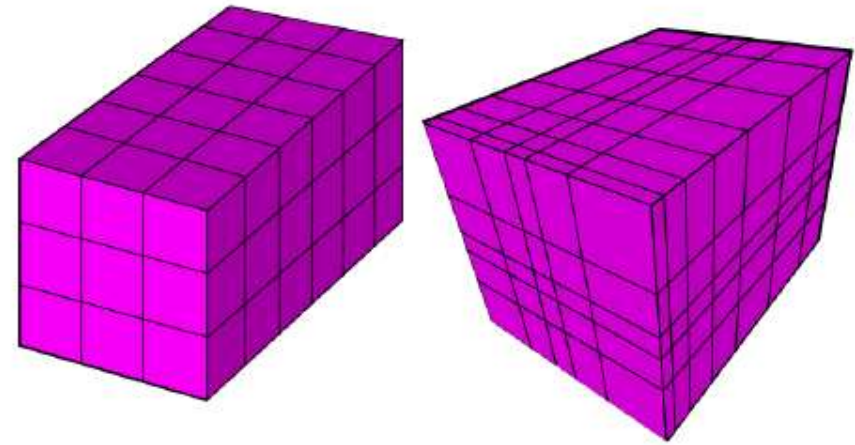
Fields: Grid Types

Uniform Grid

Geometry & topology can be computed

Rectilinear Grid

Nonuniform sampling



Information vs Scientific Visualization

- **Information Vis**

- “Abstract Data”
- Tables, Graphs
- Free to choose spatial layout

- **Scientific Vis**

- “Spatial Data” (Fields)
- Not free to choose spatial layout
- Find best way to depict reality

Other Collections

Sets

Unique items, unordered

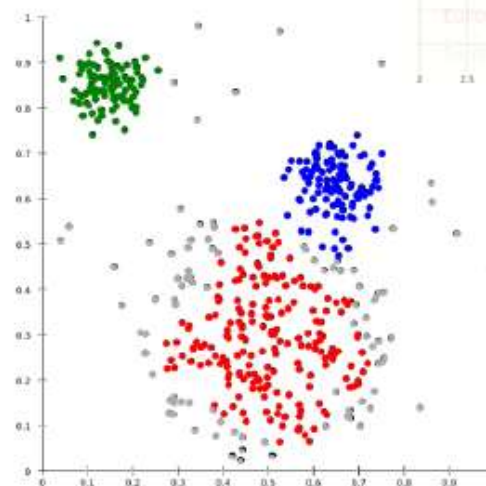
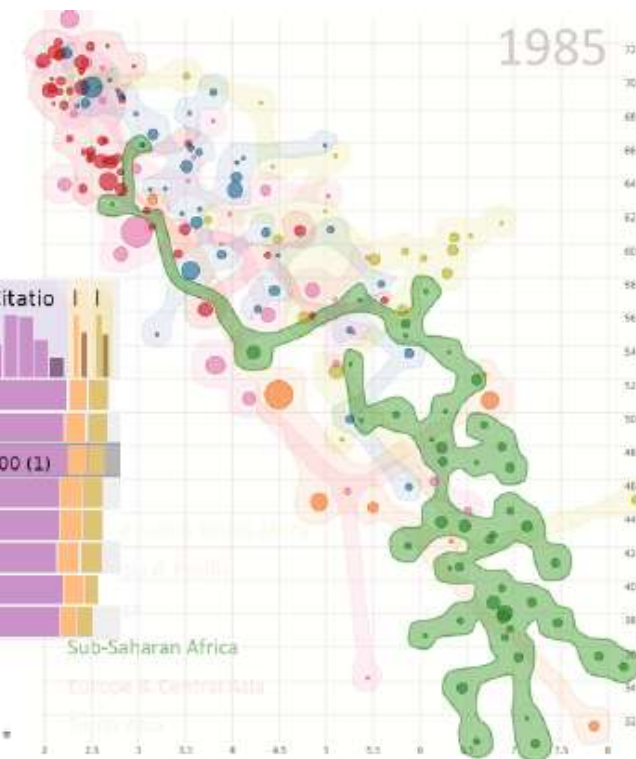
Lists

Ordered, duplicates allowed

Clusters

Groups of similar items

Rank	School Name	Academic repu	E	Facult	Citatio
	Filter: <None>				
1.	Massachusetts Inst				
2.	University of Camb				
3.	Harvard University	100 (1)			100 (1)
4.	UCL (University Co				
5.	University of Oxfor				
6.	Imperial College L				
7.	Yale University				
8.	University of Chic				



Data vs Conceptual Model

Data Model: Low-level description of the data

Set with operations, e.g., floats with +, -, /, *

Conceptual Model: Mental construction

Includes semantics, supports reasoning

Data	Conceptual
1D floats	temperature
3D vector of floats	space

Data Model -> Data Type

- From data model...
 - 32.5, 54.0, -17.3, ... (floats)
- Using conceptual model...
 - Temperature
- To data type
 - Continuous to 4 significant digits (Q)
 - Hot, warm, cold (O)
 - Burned vs. Not burned (N)

Structured vs Unstructured

- Structured Data
 - Known data types, semantics
- Unstructured Data
 - No predefined data model
 - Text-heavy, interspersed with facts (dates, times, locations)
 - Video, images
 - Need to be converted to structured data for visualization
- Unstructured -> Structured
 - Natural Language Processing
 - Text mining (sentiment, keywords, concepts, categories)

Combinations, Derived Data

- Networks can have attributes
 - Attributes have hierarchies
 - Data types can be transformed
-
- Real life is complicated...

Data Cleansing

- Data may be missing/corrupted
 - Remove?
 - Modify?
- You may want to adjust values
 - Use inverse
 - Map nominal to ordinal/quantitative
 - Normalize values
 - Scale between 0 and 1

Publicly Available Datasets

- <https://www.kaggle.com/datasets>
- <http://www.kdnuggets.com/datasets/>
- <http://www.google.com/publicdata/directory>
- <http://www.data.gov/>
- *Many more!*

Reading

- Chapter 2